AYUSH MEHTA

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SUMMARY

Software Engineer with 2+ years of experience building scalable backend services, automating high-volume data workflows, and improving production performance for enterprise applications. Experienced across the full development lifecycle from design and implementation to testing, deployment, and optimization and known for delivering reliable, maintainable systems that reduce operational effort and support users at scale.

EDUCATION

Master of Science in Computer Science Expected Graduation: Dec 2026
Seattle University Seattle, WA
Courses: *Distributed Systems, Software as a Service (SAAS), Software Architecture & Design, Software Testing & Debugging*

Bachelor of Technology in Information Technology Sep 2019 – Aug 2022
MIT Arts, Design & Technology Pune, IN
Courses: *Data Structures and Algorithms, Machine Learning, Web Technologies, Artificial Intelligence, Relational Databases*

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, C#, JavaScript, TypeScript
- **Databases & Cloud:** SQL, PostgreSQL, MySQL, MongoDB, AWS, GCP, Azure
- **Specialized Areas:** REST APIs, Microservices, Debugging, Testing, UI/UX, Automation
- **Frameworks & Tools:** Flask, Django, Node.js, React.js, Git, Docker, Jenkins, Kubernetes, CI/CD, Linux/Unix
- **Soft Skills:** Communication, Problem-Solving, Cross-Team Collaboration, Attention to Detail, Adaptability, Analytical Thinking

WORK EXPERIENCE

Security Engineer Intern | Amazon Jun 2026 – Sep 2026
- Built and deployed a Python adversarial-testing framework for AI agents, replacing manual review with repeatable dynamic tests and reducing security-review time from days to minutes.
- Integrated AWS SigV4 request signing, IAM authentication, test execution, and results collection into one workflow, automating 95% of testing for cloud-hosted AI-agent environments.
- Developed dynamic adversarial test cases for unauthorized tool and API use, boundary violations, and privilege escalation, expanding AI-agent threat coverage by more than 85%.
- Validated detected issues against actual agent behavior and Amazon CloudWatch logs, reducing false-positive findings from 60% to 10% before reporting results to engineers.

Software Engineer | Resilinc.ai Sep 2022 – Aug 2024
- Led design reviews and built a Python and Flask assessment module with validated multilingual data handling, automating 1,000+ file downloads per day and reducing processing time by 60%.
- Designed and deployed a high-performance Python framework using Redis caching and asynchronous I/O, reducing error rates by 40% while scaling traffic capacity 1.5x for 10,000+ users.
- Refactored SQL queries with parameterization, indexing, and caching to reduce injection risk, cut compute time by 30%, and improve dashboard responsiveness for enterprise users.
- Enhanced a real-time supply-chain risk analytics dashboard with React and MongoDB, enforcing role-based access controls and improving decision-making efficiency by 35% for 80+ clients.

Python Developer | Resilinc.ai Mar 2022 – Aug 2022
- Developed Python scripts integrated with REST APIs to automate data workflows, filtering more than 500,000 real-time records monthly and improving data accuracy to 95% in across data workflows.
- Containerized modular microservices using Docker and applied scalable design patterns with comprehensive documentation, improving onboarding consistency and accelerating deployment across the team.
- Diagnosed high-impact web-service bottlenecks through performance profiling and SQL optimization, reducing downtime from two hours to 30 minutes and improving service reliability.
- Streamlined delivery across five departments by automating Jira, Confluence, and GitLab CI/CD workflows, improving milestone-delivery efficiency by 30% and reducing manual coordination.

PROJECTS

EchoJournal - Mood Booster Platform | Seattle University Dec 2025
- Designed and implemented backend components for a full-stack journaling platform, enabling users to securely create personal entries and track emotional well-being over time.
- Developed Node.js and Express microservices with Google SSO authentication and access-control checks, protecting user data and supporting privacy-focused application workflows.
- Collaborated with teammates to integrate MongoDB, build automated testing suites, and present the completed application, coordinating technical implementation across backend and frontend work.

Image Processing System| Seattle University Aug 2025
- Led development of a distributed image-transformation engine in Python using gRPC and Docker, enabling batch processing for grayscale, resize, and flip operations across modular services.
- Applied a pipe-and-filter architecture to separate transformation stages into reusable components, supporting deployment across command-line, web, and mobile application environments.
- Assessed performance, memory efficiency, and observability requirements while designing the distributed workflow, documenting technical trade-offs to support future feature extensions.